NLP

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1) What is NLP?

→ It is a field of AI that enables computers to understand & interpret & generate human language

It combines linguistics + machine learning + DL

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2) Types of NLP

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- Natural Language Understanding (NLU)
  * Focuses on understanding meaning
  * Example : chatbot understanding a question
- Natural language Generating (NLG)
  * Focuses on generating text
  * Example : ChatGPT generating answer

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3) Challenges in NLP

→ * Ambiguity
- Same word, different meaning or multiple
  eg: "bank" (river bank / money bank)

* Context understanding
- Meaning changes based on sentence

* Sarcasm & sentiment
- "Great job!"

* Language diversity
- Different languages, dialects

* Data limitations
- Lack of labeled data

4) Difference NLP vs NLU, NS NLG

NLP	NLU	NLG
Complete field	Understanding	Generating
Includes NLU & NLG	Extract meaning	Produce text
Eg: Translation	Intent detection	Text generation

5) Techniques of NLP

- Tokenization

Breaking text into words

eg: Deep learning is powerful

- Stop word Removal

Removing common words

- Stemming

Reducing words to root

- Lemmatization

converting to meaningful base form

POS Tagging

Identifies word types

Eg: She runs fast

She → pronoun

runs → verb

fast → Adverb

Named Entity Recognition

Identifies entities

Eg : Elon Musk founded Tesla

Elon Musk → person

Tesla → organization

• Bag of words (BoW)

Counts word frequency

TF - IDF

Measures word importance

$$TF = \frac{\text{word count}}{\text{total words}}$$

$$IDF = \log\left(\frac{\text{total documents}}{\text{documents containing word}}\right)$$

• Word2Vec

Converts words into vectors

• Examples

king - man + women ≈ queen

• N-gram

Groups of words

Deep learning is fun

• Bigram ["Deep learning" "learning is"]